

Star Formation and Black Hole activity at Cosmic Dawn: A JADES Spectroscopic Investigation

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Abstract

Historically, active galactic nuclei (AGN) and star-forming (SF) galaxies have often been studied as distinct populations. Yet, it remains unclear how early supermassive black holes formed and whether their activity influenced star formation in young galaxies in the early universe. To investigate this relationship, a sample of high-redshift galaxies was selected from JADES, including galaxies with both AGN and SF characteristics. Emission-line ratios and spectral line widths were analyzed and revealed that some galaxies display both star-forming and AGN-like characteristics. While FWHM measurements alone did not provide a clear distinction between the selected populations, the results suggest that black hole growth and star formation may coexist in some galaxies during cosmic dawn.

Introduction

The universe is continuously expanding and there is so much to learn about its earliest stages, a period known as the cosmic dawn. The cosmic dawn generally refers to the first billion years of the universe's history, corresponding to redshifts of $z \gtrsim 5$. During this time, the first generations of galaxies were forming, stars were being born at high rates, and supermassive black holes may have been already growing in galaxy centers. Yet, it remains unclear how these early black holes formed and whether their activity influenced star formation in young galaxies. This raises an important question: how are active galactic nuclei (AGN) and star formation (SF) connected during the cosmic dawn?

The JWST Advanced Deep Extragalactic Survey (JADES) provides ideal datasets for investigating this question. JADES is a major survey designed to explore the universe's first stars and galaxies through deep-field observations. One of the central goals of modern astronomy is understanding how galaxies evolved over cosmic time, particularly the relationship between early star formation and black hole growth during the first billion years after the big bang. This project investigates how spectral features in JADES data can be used to distinguish between actively star-forming galaxies and those hosting active black holes at high redshift.

Distinguishing between SF galaxies and AGN is possible because both produce strong emission lines, but the physical source of those lines are different. Star-forming galaxies contain large amounts of ionized gas produced by their surplus of young hot stars, resulting in strong hydrogen recombination lines such as $H\alpha$ and $H\beta$ that are typically narrow. AGN emission lines are produced by ionized gas surrounding accreting supermassive black holes. Because of this intense radiation generated by accretion, stronger ionization lines are produced, resulting in elevated $\frac{[OIII]}{H\beta}$ ratios compared to typical SF galaxies. This ratio serves as a measure of the ionization strength within a galaxy and is commonly used to distinguish AGN activity from star formation.

However, elevated $\frac{[OIII]}{H\beta}$ ratios have been shown to be common among high-redshift galaxies and are often associated with harder ionizing radiation fields compared to local galaxy populations (Sanders et al. 2023). AGN emission lines may also appear broadened due to the high velocities of gas near the central black hole.

JADES data have been released in phases, each covering a specific region of sky or type of observation. The survey uses three main instruments: the Near-Infrared Camera (NIRCam) for deep imaging, the Near-Infrared Spectrograph (NIRSpec) for multi-object spectroscopy, and additional data from the Mid-Infrared Instrument (MIRI). The dataset analyzed is part of JADES GOODS-South NIRSpec prism spectroscopy products available through MAST, including the v1.1 line-flux catalog and corresponding clear prism x1d spectra. These observations probe extremely distant galaxies whose light was emitted when the universe was less than 1 billion years old.

The raw JWST data undergo standard calibration procedures before scientific analysis, including dark current subtraction, flat field correction, cosmic ray removal, background subtraction, wavelength calibration, and flux calibration. The two-dimensional spectral images are then extracted into one-dimensional spectra corresponding to individual galaxies. These calibration procedures produce science-ready data suitable for analysis.

Galaxies were selected from the JADES GOODS-South NIRSpec line-flux catalog by requiring secure spectroscopic redshift flags of A, B or C, along with redshifts $z \geq 5$. Flags A and B represent highly robust spectroscopic redshift measurements identified from strong emission lines, while flag C represents secure redshifts determined from spectral breaks and/or lower signal-to-noise emission features. Emission lines fluxes for $H\alpha$, $H\beta$, and $[OIII]$, were then used to identify candidate star-forming and AGN-host galaxies through their spectral line properties and $\frac{[OIII]}{H\beta}$ ratios.

Emission-line strengths and widths were analyzed to identify signatures of star formation and potential AGN activity within a sample of high-redshift JADES galaxies. By comparing ionization diagnostics and spectral properties, this investigation explores the relationship between black hole activity and star formation during early galaxy evolution.

Data Analysis

The primary catalog used in this investigation was the JADES Data Release 3 (DR3.1.3) NIRSpec line-flux catalog (D'Eugenio et al. 2024), which contains measured emission-line fluxes and spectroscopic redshifts for galaxies in the GOODS- South field.

After loading the JADES GOODS-South NIRSpec line-flux catalog into Python using the `astropy.io.fits` and `pandas` libraries, I filtered the dataset to include only galaxies with secure spectroscopic redshift flags (A, B, or C) and redshifts greater than $z > 5$. From the catalog, columns containing galaxy coordinates, redshifts, and emission line fluxes for $H\alpha$ 6563, $H\beta$ 4861, $[OIII]$ 5007, and $[OII]$ 3727 were extracted, as shown in Table 1.

Table 1: Selected high-redshift galaxy candidates, including galaxy identification, redshift measurements and spectra quantities. *approximately based off redshifts

NIRSpec ID	RA	Dec	Z	Flag	$H\alpha$ flux	$H\beta$ flux	$[OIII]$ flux	$\frac{[OIII]}{H\beta}$ ratio	Universe age*
4297	53.156	-27.815	6.71	A	1.06 ± 0.04	0.38 ± 0.03	2.43 ± 0.08	6.30 ± 0.47	0.8
4873	53.166	-27.786	5.46	C	6.54 ± 0.56	1.41 ± 0.90	13.95 ± 0.87	9.86 ± 6.30	1.0
2430	53.128	-27.788	5.49	B	1.92 ± 0.11	0.73 ± 0.12	2.80 ± 0.13	3.83 ± 0.69	1.0
4404	53.115	-27.815	5.76	A	2.36 ± 0.04	0.95 ± 0.04	4.41 ± 0.04	4.65 ± 0.19	0.95

To identify candidate star-forming and AGN-host galaxies, the $[OIII]/H\beta$ emission line ratio for each galaxy was calculated by dividing the $[OIII]$ 5007 flux by the $H\beta$ 4861 flux. The uncertainty on this ratio was calculated using standard error propagation:

$\sigma_R = R \sqrt{\left(\frac{\sigma_{[OIII]}}{[OIII]}\right)^2 + \left(\frac{\sigma_{H\beta}}{H\beta}\right)^2}$, where $R = \frac{[OIII]}{H\beta}$ ratio, $\sigma_{[OIII]}$ = error on $[OIII]$ flux and $\sigma_{H\beta}$ = error on $H\beta$ flux. A threshold of $\frac{[OIII]}{H\beta} > 5$ was adopted to identify galaxies with substantially elevated ionization relative to the comparison sample, while galaxies with lower ratios and strong $H\alpha$ emission were classified as star-forming candidates.

The $\frac{[OIII]}{H\beta}$ ratios were plotted as a function of redshift to investigate how ionization strength varied across the selected sample. Different marker shapes were additionally used to represent the JADES spectroscopic quality flags (A, B, and C).

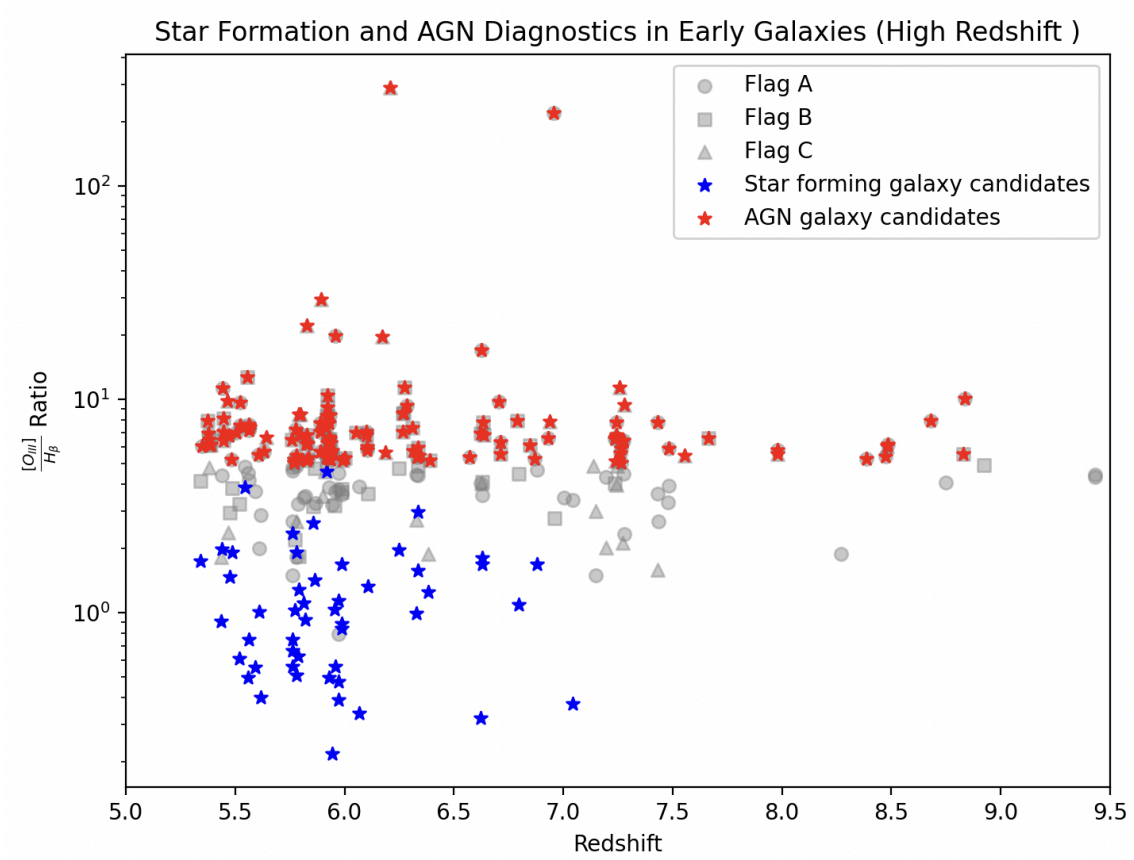


Figure 1: $\frac{[OIII]}{H\beta}$ ratios as a function of redshift for galaxies in JADES GOODS-South catalog.

For several selected candidate galaxies, one-dimensional JWST/NIRSpec prism spectra were retrieved directly from the Mikulski Archive for Space Telescopes (MAST) using the astroquery.mast package. The downloaded x1d.fits files contained wavelength, flux, and flux-uncertainty measurements for each galaxy.

To directly compare galaxies at different redshifts, the observed wavelengths were converted into rest-frame wavelengths using : $\lambda_{rest} = \frac{\lambda_{obs}}{z+1}$. The rest-frame spectra were then plotted and compared to identify differences in emission-line strength and ionization properties between the selected galaxies.

Because AGN can produce broadened emission lines from rapidly moving gas near the central black hole, FWHM measurements were used as an additional diagnostic beyond emission-line ratios. In order to estimate the velocity broadening of the emission lines, the full width at half maximum (FWHM) was measured for the strongest spectral features in each galaxy. The wavelength range was restricted (0.45-0.70 μm) which contains the $H\beta$, $[OIII]$ and $H\alpha$ emission lines. Peaks were identified using a peak-finding routine adapted from code developed by Williams and Hedge. Candidate peaks were required to have a higher flux than neighboring

pixels and exceed a manually selected flux threshold. For each detected peak, the half-max value was calculated by dividing the peak flux by two. The code then searched to the left and right of the peak to find the nearest wavelength points where the flux dropped below these half-max values. The two selected wavelengths were used as FWHM boundaries, as shown on Figures 3-6. The FWHM was calculated as the difference between the right and left boundary wavelengths. A flux-weighted centroid was also calculated for each peak, using the wavelength and flux values within the FWHM boundaries.

Results

As shown in Figure 1, the diagnostic plots produced from the JADES GOODS-South sample show a noticeable separation between the selected star-forming and AGN-like galaxy candidates. Galaxies classified as AGN-like candidates generally exhibit elevated $\frac{[OIII]}{H\beta}$ ratios compared to the star-forming comparison sample, suggesting the presence of harder ionizing radiation fields. In contrast, the star-forming galaxies tend to cluster at lower $\frac{[OIII]}{H\beta}$ ratios, consistent with ionization dominated by young massive stars rather than accreting black holes. While the sample size is limited, the separation suggests that $\frac{[OIII]}{H\beta}$ remains a useful diagnostic for identifying galaxies with enhanced ionization at high redshift.

Several individual galaxies also stood out during the analysis. This is reflected in Figure 2, where Galaxy 4297 showed one of the strongest $\frac{[OIII]}{H\beta}$ ratios in the selected sample, making it one of the strongest AGN-like candidates. Galaxy 4873 displayed both strong $H\alpha$ emission and elevated oxygen emission lines, suggesting that active star formation and enhanced ionization may be occurring simultaneously. Galaxies 2430 and 4404 showed lower ionization ratios while still maintaining strong hydrogen emission features, making them more consistent with typical star-forming galaxies.

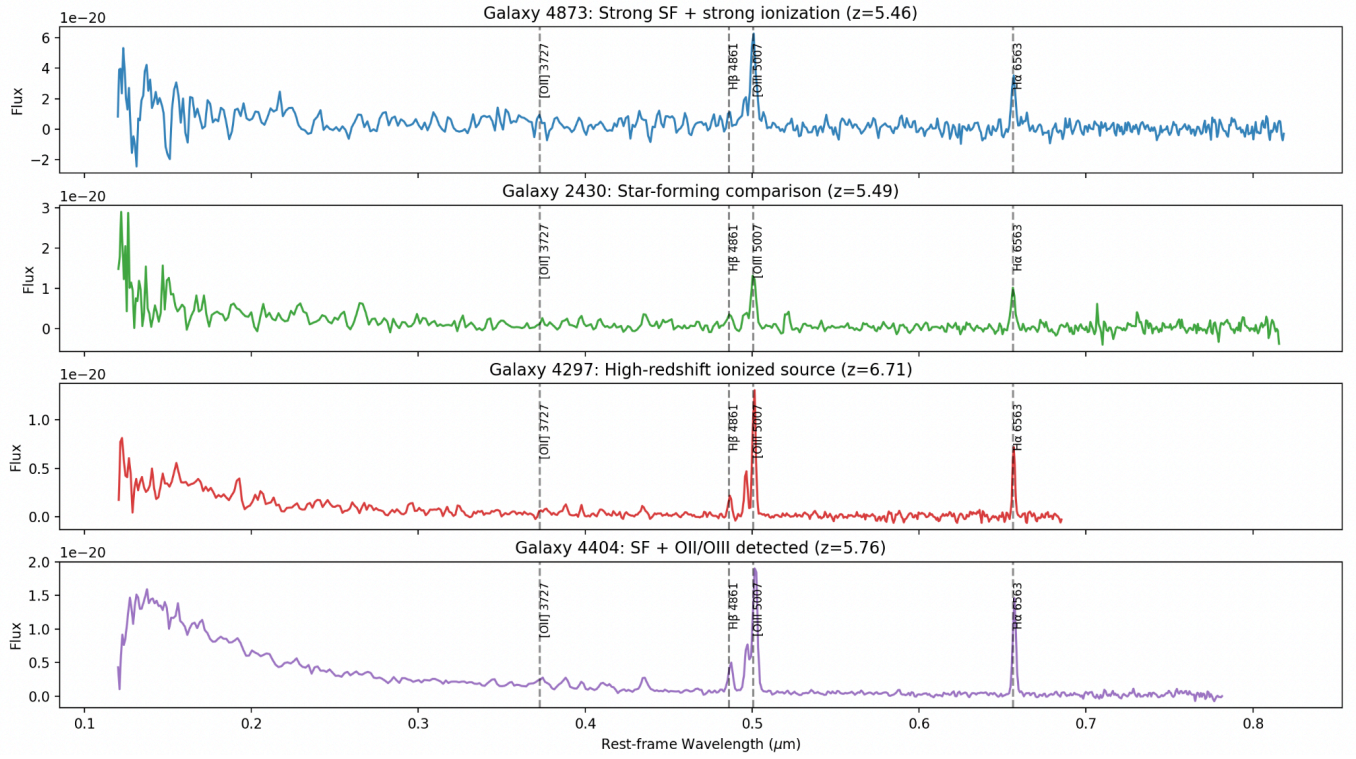
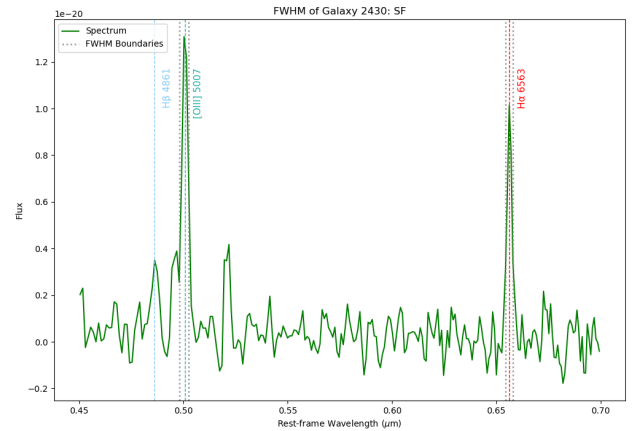
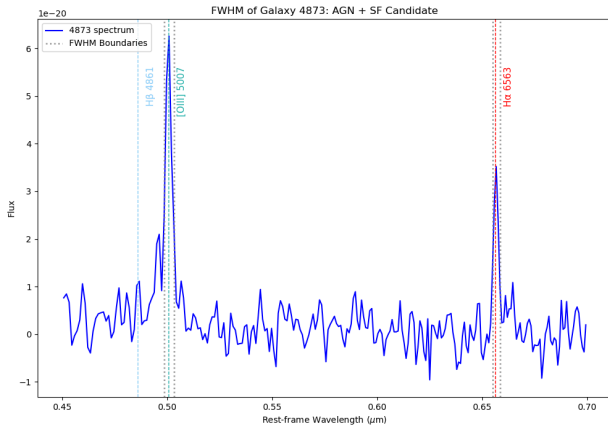
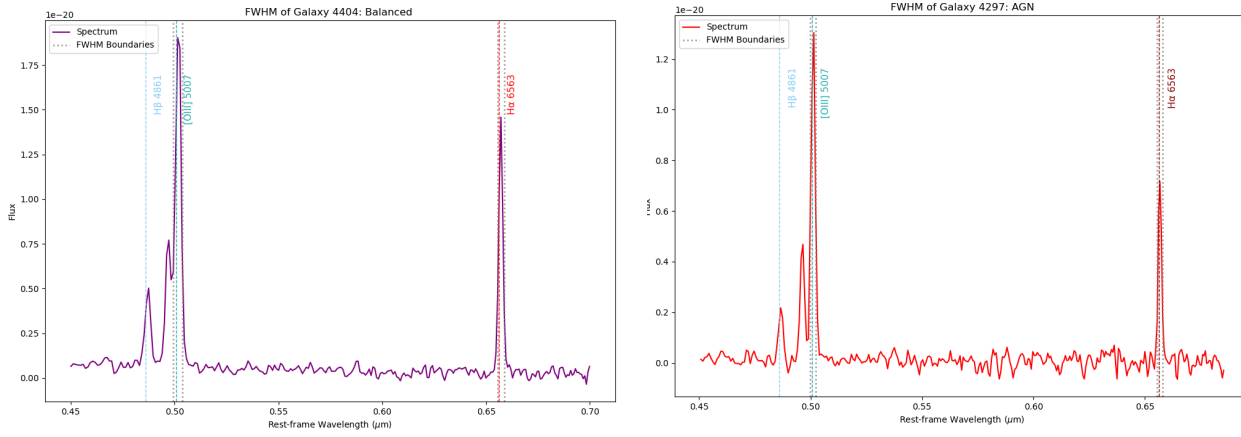


Figure 2 : Rest-frame spectra of the 4 candidates chosen, with major emission features being highlighted.

The rest-frame JWST/NIRSpec prism spectra additionally revealed visible differences in emission line structure and relative line strength between the selected galaxies. After converting the spectra into rest-frame wavelength space, the major emission features, including $H\alpha$, $H\beta$, and $[OIII]$, aligned consistently across galaxies despite their differing redshifts. The AGN-like candidates generally exhibited stronger oxygen emission relative to the hydrogen recombination lines than the star-forming comparison galaxies, consistent with elevated $\frac{[OIII]}{H\beta}$ ratios measured from the catalog.





Figures 3-6: Each figure's color corresponds to the spectra shown above in Figure 2. 3 main peaks are shown, and were fitted with FWHM. Also shown, the emission lines and their rest positions.

Finally, each measured FWHM was converted into a velocity width using $v = c * (\Delta\lambda / \lambda_0)$, where $\Delta\lambda$ is the FWHM, λ_0 is the measured centroid wavelength, and c is the speed of light. These velocity widths were then compared across the selected galaxies to test whether AGN-like candidates showed broad emission line features against the SF galaxies.

Table 2: Velocity widths derived from FWHM measurements.

Galaxy ID	[OIII] Velocity (km/s)	H α Velocity (km/s)
4873 (AGN + SF Candidate)	2786.1	1609.7
2430 (SF)	2771.9	1600.8
4297 (AGN)	1464.1	1114.4
4404 (Balanced)	2539.1	1465.0

After comparison, it was found that the measured velocity widths ranged from 1100-2800 km/s. However, comparable velocities were observed in both the star forming and AGN candidates. In particular, Galaxy 4297, one of the strongest AGN candidates based on its elevated $\frac{[OIII]}{H\beta}$ ratio, did not exhibit the largest measured velocity widths. Instead similar or larger widths were observed in galaxies classified as star-forming. This suggests that the measured line widths are likely influenced by the instrumental resolution, line blending, and the limited spectral resolution of the prism observations. As a result, FWHM measurements alone were not sufficient to clearly distinguish between the selected galaxy populations.

Upon further spectral analysis, the selected galaxies do not separate cleanly into pure star forming and AGN populations. While some galaxies display characteristics more consistent with

one category, galaxy 4873 exhibits strong SF signatures and AGN-like ionization. The inability of line widths and traditional diagnostic methods to provide a definitive classification highlights the challenges of identifying AGN in high-redshift galaxies. This suggests that black hole growth and star formation may coexist in some high redshift galaxies, and highlights the complexity of galaxy evolution during the cosmic dawn.

Strengths and Limitations

One strength of this investigation is that it uses real JWST spectroscopic observations of galaxies from the first billion years of cosmic history. The JADES dataset provides high-quality infrared spectra capable of probing emission lines in extremely distant galaxies that were previously difficult to observe before JWST. Another strength of the project is the combination of catalog-based line diagnostics with direct analysis of retrieved NIRSpec prism spectra, allowing both quantitative and visual comparison of galaxy properties.

However, this investigation also contains several limitations. The sample size analyzed in detail is relatively small, meaning that broader statistical conclusions about the full high-redshift galaxy population cannot yet be made. Additionally, only four galaxies were analyzed in detail through direct spectral comparison and FWHM measurements, limiting the statistical significance of the results. The project also primarily relies on published catalog line measurements rather than independently fitting all spectral emission lines directly from the raw spectra. As a result, the AGN classifications presented in this investigation should be considered AGN-like candidates rather than confirmed AGN detections.

Connections to Galaxy Evolution

This investigation connects strongly to the broader field of galaxy evolution because it explores the relationship between early star formation and supermassive black hole growth during the cosmic dawn. Modern galaxy evolution models suggest that galaxies and their central black holes evolve together over cosmic time, but it is still uncertain how early these black holes formed or how strongly they influenced surrounding star formation. The mixed AGN and star-forming signatures observed in galaxies such as 4873 suggest that black hole growth and star formation may occur simultaneously during early galaxy evolution. By identifying galaxies that display possible AGN signatures at very high redshift, this project contributes to understanding how quickly supermassive black holes may have formed in the early universe.

The results additionally relate to larger questions about feedback processes in galaxy evolution. AGN activity can inject large amounts of energy into surrounding gas, potentially either suppressing or enhancing star formation within galaxies. Studying these processes in early galaxies helps astronomers better understand how the first massive galaxies evolved into the galaxy populations observed in the present-day universe.

Importance to Society

This investigation is important beyond astronomy because it explores some of the earliest stages of the universe's history and humanity's cosmic origins. The galaxies studied in this project existed less than one billion years after the Big Bang, meaning their light provides a direct view into the universe's earliest generations of stars, galaxies, and black holes. Understanding how these early systems formed helps explain how the large-scale structure of the universe developed over billions of years, eventually leading to galaxies such as the Milky Way.

This project also demonstrates the scientific importance of the James Webb Space Telescope and modern astronomical data analysis. JWST allows scientists to observe farther back in time than ever before, producing enormous amounts of publicly available scientific data that can be analyzed by students and researchers alike. Investigations such as this one help make cutting-edge astrophysical discoveries more accessible while also contributing to our broader understanding of the universe and humanity's place within it.

Response to Peer Review

Several revisions were made to this project in response to peer feedback. Additional background information was added to explain the significance of the $\frac{[OIII]}{H\beta}$ ratio, and its role in distinguishing between SF and AGN galaxies. The discussion of the JADES dataset was expanded to include more detail about the data products and selection criteria used in analysis. In addition, the introduction and several sections throughout the paper were revised for clarity, organization and overall helped strengthen the scientific context of the project.

References

- D'Eugenio, F., Cameron, A. J., Scholtz, J., Carniani, S., Willott, C. J., et al. 2024, *JADES Data Release 3 – NIRSpec/MSA spectroscopy for 4,000 galaxies in the GOODS fields*, submitted to *The Astrophysical Journal Supplement Series*.
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